

Institute of Actuaries of India

Subject CM2A – Financial Engineering and Loss Reserving (Paper A)

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i) Answer is c. [1]

$$E[A] = (.20 \cdot .3 + .04 \cdot .2 + .10 \cdot .5) = 11.8\%$$

$$E[B] = (.02 \cdot .3 + .10 \cdot .2 + .06 \cdot .5) = 5.6\%$$

ii) Answer is a. [2]

$$S.D.[A] = [(.20^2 \cdot .3 + .04^2 \cdot .2 + .10^2 \cdot .5) - .118^2]^{0.5} = 5.83\%$$

$$S.D.[B] = [(.02^2 \cdot .3 + .10^2 \cdot .2 + .06^2 \cdot .5) - .056^2]^{0.5} = 2.80\%$$

iii) Answer is b [1]

$$\rho_{AB} = \frac{COV(A,B)}{\sigma_A \sigma_B}$$

$$COV(A, B) = E[AB] - E[A] E[B]$$

$$E[AB] = 0.5\%$$

$$COV(A, B) = 0.5\% - 11.8\% \cdot 5.6\%$$

$$COV(A, B) = -0.1608\%$$

$$\rho_{AB} = \frac{-0.1608\%}{5.83\% \cdot 2.80\%}$$

$$\rho_{AB} = -0.9855$$

[1]

iv) This shows that the returns are strongly negatively correlated, so a poor return on one asset is very likely to be accompanied by a favourable return on the other. [1]

[6 Marks]**Solution 2:**

Analysts and investment managers can be unreasonably drawn to a stock based on understanding and belief in the company's business model and/or management. This affection can lead them to persistently discount bad news about the stock.

Investment managers can be highly confident in their own abilities. Over confidence can lead the practice of managers using good news on a stock to reinforce their position whilst putting any bad news down to the ignorance of other market participants.

Managers will be reticent to take responsibility and crystallize a loss having defended the position to date.

[5 Marks]**Solution 3:**

i) Interval:

Required interval is where $x+1 < \exp(-c \cdot x)$ for positive x . Note that:

- $x+1 \leq 1 \leq \exp(-c \cdot x)$ when x lies in $[-1, 0]$
- $\exp(-c \cdot x) \leq 1 \leq x+1$ when x lies in $[0, \infty)$

So, the answer is $[-1, 0]$.

[1]

ii) Value of c:

For $f(x)$ to be a valid PDF, we require: $\int_{-1}^{\infty} f(x)dx = 1$

This can be broken as: $\int_{-1}^0 (x+1)dx + \int_0^{\infty} e^{-cx}dx = 1$

Evaluating the integrals and simplifying yields: $1/2 + 1/c = 1$. Therefore, $c = 2$.

[2]

iii) Expected return:

Expected return can be obtained as $E[X] = \int_{-1}^{\infty} xf(x)dx$.

Breaking down the integral, we get: $E[X] = \int_{-1}^0 x(x+1)dx + \int_0^{\infty} xe^{-2x}dx$

Simplifying the first integral: $\int_{-1}^0 x(x+1)dx = -\frac{1}{6}$

Simplifying the second integral: $\int_0^{\infty} xe^{-2x}dx = \frac{1}{4}$

Thus, $E[X] = -\frac{1}{6} + \frac{1}{4} = \frac{1}{12}$. In other words, the expected return is 8.33%.

[3]

iv) $E[X^2]$:

As before, we split the integral in two terms: $E[X^2] = \int_{-1}^0 x^2(x+1)dx + \int_0^{\infty} x^2e^{-2x}dx$

Simplifying the first integral: $\int_{-1}^0 x^2(x+1)dx = \frac{1}{12}$

Simplifying the second integral: $\int_0^{\infty} x^2e^{-2x}dx = \frac{1}{4}$

So, $E[X^2] = 1/12 + 1/4 = 1/3$

[3]

v) Variance

Variance can be found as $E[(X - \mu)^2] = E(X^2) - E(X)^2$

Thus, variance is $\frac{1}{12} + \frac{1}{4} - \left(\frac{1}{12}\right)^2 = \frac{47}{144}$

[1]

vi) 90% VaR indicates the maximum level of loss with a 90% confidence, i.e. there is a 10% probability of a greater loss. [0.5]

We start by finding t , where $P(X < t) = 0.1$

[0.5]

Thus, $0.1 = \int_{-1}^t (x+1)dx = \left[\frac{x^2}{2} + x\right]_{-1}^t = \frac{t^2}{2} + t + \frac{1}{2}$

[0.5]

Simplifying yields $t^2 + 2t + 0.8 = 0$. Solving for t and rejecting the roots outside $[-1, 0]$, we are left with $t = -0.5527864...$ (i.e. about 55.28% loss). [1]

Therefore, 90% VaR for a \$1m investment is $\$1m \cdot 0.5527864... = \$552,786$.

[0.5]

[3]

vii) Shortfall Probability (SP) / Expected Shortfall (ES) at -50% threshold:

For a random variable X and a threshold level K ,

Shortfall Probability = $\Pr(X < K)$ and

Expected Shortfall = $E[\max(K - X, 0)]$

[1]

Therefore, in this instance, $SP = \int_{-1}^{-0.5} (x + 1)dx = \frac{1}{8}$ [1]

Similarly, $ES = \int_{-1}^{-0.5} (-0.5 - x)(x + 1)dx = \frac{1}{48}$.

Thus, the expected loss below -50% threshold for a \$1m investment will be $\$1m/48 = \$20,833$. [1]

Use of a shortfall risk measure corresponds to a utility function that has a discontinuity at the minimum required return (threshold level). [1]

[4]

[17 Marks]

Solution 4:

i) Filling in the missing entries

Put-call parity states that: $c + Ke^{-rt} = p + S$ [0.5]

Each row in the table corresponds to a difference value of K (and hence c and p) while S, r, t remain unchanged. While t is known to be 6 months (= 0.5 years), S and r remain to be solved for.

Using the second row, we get: $1342.92 + 11000 * e^{-0.5r} = 625.11 + S$

Using the third row, we get: $856.82 + 12000 * e^{-0.5r} = 1119.21 + S$ [1]

Solving these two equation simultaneously for S and r yields: $S = 11500$ and $r = 4\%$ [1.5]

The missing entry (p for K = 10000) can be found by writing the put-call parity for first two:

$$1990.88 + 10000 * e^{-0.04 * 0.5} = p + 11500$$

This gives $p = 292.87$ [1]

[4]

ii) ACTEX Future price:

The fair price for the 6-month future contract as set today should be: $Se^{rt} = 11732.32$ [1]

iii) Resilience Future price:

The present value for the dividend income is $I = 6e^{-0.04 * (2/12)} = 5.96$ [1]

The fair price for the 6-month future contract as set today should be: $(S-I)e^{rt} = 197.96$ [1]

[2]

iv) Assumptions - Binomial model:

In the binomial model, it is assumed that:

- There are no taxes or trading (=transaction) costs [0.5]
- Assets (stocks and bonds) can only be bought and sold at discrete times [0.5]
- Assets may be held in any amounts (i.e. no minimum or maximum units of trading) [0.5]
- Principle of no arbitrage applied (i.e. there are no arbitrage opportunities) [0.5]

[2]

v) Value of 1-month put option on Resilience:

The current Resilience stock price is \$200. One month from now, the possible prices and the corresponding put option pay-offs are as follows:

Scenario	Stock price	Put option payoff
Up (u)	210	$0 = \max\{205-210, 0\}$
Down (d)	195	$10 = \max\{205-195, 0\}$

[1]

Let's hold s units of stock and c units of cash to replicate the put option payoff. As the current value of stock is \$200, the current value of this portfolio is: $200s + c$.

[0.5]

After 1 month, the cash will grow at the risk free rate r and the portfolio will have the value:

Scenario	Stock price	Portfolio value
Up (u)	210	$210s + c(e^{0.04/12})$
Down (d)	195	$195s + c(e^{0.04/12})$

[1]

Equating the option payoff under the two scenarios with the corresponding portfolio value gives a set of simultaneous equations: $210s + c(e^{0.04/12}) = 0$; $195s + c(e^{0.04/12}) = 10$.

[0.5]

Solving these gives: $s = -2/3$ and $c = \$139.53$.

[1]

Therefore the current value of the portfolio is: $200s + c = \$6.20$

[0.5]

By the principle of no arbitrage and the law of one price, since our portfolio has the same value as the payoff of the derivative one month from now (under all scenarios), the current price of the derivative should be the same as the current value of the portfolio, i.e. \$6.20.

[0.5]

[5]

vi) Value of long future contract on Growth Industries:

The current value of a long position of a forward contract on Growth Industries is given as:

$$f = S - Ke^{-r(T-t)}$$

[S = current stock price; K = strike price; r = (continuously compounded risk-free annual) rate of interest; t = time to expiry]

[1]

vii) Greeks for Black Scholes PDE:

Differentiating f w.r.t. t ,

$$\Theta = -rKe^{-r(T-t)}$$

[1]

Differentiating f w.r.t. S ,

$$\Delta = 1$$

[0.5]

Differentiating Δ w.r.t. S ,

$$\Gamma = 0$$

[0.5]

[2]

viii) Boundary Condition:

At expiry ($t = T$),

$$f = S_T - K$$

[0.5]

This is expected as the long position holder will receive the stock (worth S_T) and have to pay K in cash.

[0.5]

[1]

[18 Marks]

Solution 5:

i) Answer is b

[2]

ii) Answer is a

[2]

iii) Parameters for valuing the call option payoff using the black scholes formula are as follows:

$$F(0) = 30 \quad [1]$$

$$K=10$$

$$T-t = 10$$

$$\sigma = 0.3$$

$$r = .06$$

$$d1 = [\ln(30/10) + (.06 + (0.3^2/2)) * 10] / (0.3 * 10^{(1/2)})$$

$$d1 = 2.2648 \quad [1]$$

$$d2 = d1 - (.3 * 10^{(1/2)}) = 1.3161 \quad [0.5]$$

$$\varphi(d1) = 0.98824 \quad [0.5]$$

$$\varphi(d2) = 0.90593 \quad [0.5]$$

$$E(0) = 30 * 0.98824 - 10 * e^{-.06*10} * .90593$$

$$E(0) = 24.6754 \quad [1]$$

$$B(0) = F(0) - E(0) = 5.3246 \quad [0.5]$$

$$[5]$$

[9 Marks]

Solution 6:

i) Answer is False. [1]

The log normal model assumes that the returns over non-overlapping intervals are independent.

ii) Answer is c. [3]

$$\text{Log } S_2 - \text{Log } S_0 \sim N(16\%, 12.5\%)$$

$$E(S_2 / S_0) = S_0 e^{(\mu + \frac{\sigma^2}{2})} = 1200 * e^{(.16 + .125/2)} = 1200 * 1.2492 = 1499.04$$

$$V(S_2 / S_0) = (S_0)^2 e^{(2\mu + \sigma^2)} (e^{\sigma^2} - 1) = 1200^2 * e^{(2*.16 + .125)} * (e^{.125} - 1) = 1200^2 * 1.56049 * .13315 = 299202 = 547^2 \text{ approx.}$$

[4 Marks]

Solution 7:

i)

$$dr_t = 0.8(0.04 - r_t) dt + 0.1 dB_t$$

Using a Taylor series expansion for the given function

$$d(r_t e^{0.8t}) = df(r_t, t)$$

$$= \frac{\partial f}{\partial r_t} dr_t + \frac{1}{2} \frac{\partial^2 f}{\partial r_t^2} (dr_t)^2 + \frac{\partial f}{\partial t} dt \quad - (i)$$

$$\frac{\partial f}{\partial r_t} = e^{0.8t}$$

$$\frac{\partial^2 f}{\partial r_t^2} = 0$$

$$\frac{\partial f}{\partial t} = 0.8e^{0.8t} r_t \quad [1]$$

$$dr_t = 0.8(0.04 - r_t) dt + 0.1 dB_t$$

Substituting the above values in equation (i) we get,

$$\begin{aligned} &= e^{0.8t} [0.8(0.04 - r_t) dt + 0.1 dB_t] + 0.8e^{0.8t} r_t dt \\ &= 0.032e^{0.8t} dt + 0.1e^{0.8t} dB_t \end{aligned} \quad [1]$$

Substituting s for t and integrating between 0 and t , we get:

$$\int_0^t d(r_s e^{0.8s}) = \int_0^t 0.032e^{0.8s} ds + \int_0^t 0.1e^{0.8s} dB_s \quad [1]$$

$$[r_s e^{0.8s}]_0^t = 0.04 [e^{0.8s}]_0^t + 0.1 \int_0^t e^{0.8s} dB_s$$

$$r_t e^{0.8t} - r_0 = 0.04 (e^{0.8t} - 1) + 0.1 \int_0^t e^{0.8s} dB_s \quad [1]$$

$$r_t = r_0 e^{-0.8t} + 0.04 (1 - e^{-0.8t}) + 0.1 \int_0^t e^{0.8(s-t)} dB_s$$

$$r_t = .04 + e^{-0.8t} (r_0 - 0.04) + 0.1 \int_0^t e^{0.8(s-t)} dB_s \quad [1]$$

[5]

ii)

$$dB_s \sim N(0, ds)$$

$$\sigma e^{0.8(s-t)} dB_s \sim N(0, 0.1^2 e^{1.6(s-t)} ds)$$

$$\int_0^t \sigma e^{a(s-t)} dB_s \sim N(0, \int_0^t 0.1^2 e^{1.6(s-t)} ds) \quad [1]$$

$$r_t \sim N\{0.04 + e^{-0.8t} (r_0 - 0.04), \int_0^t 0.1^2 e^{1.6(s-t)} ds\} \quad [1]$$

$$r_t \sim N\{0.04 + e^{-0.8t} (r_0 - 0.04), \frac{0.1^2}{1.6} (1 - e^{-1.6t})\} \quad [1]$$

$$\text{As } t \rightarrow \infty, e^{-1.6t} \rightarrow 0$$

$$r_t \sim N(0.04, 0.00625) \quad [1]$$

[4]

iii)

$$E[r_t | F_s] = E[.04 + e^{-0.8t} (r_0 - .04) + \int_0^t 0.1e^{0.8(u-t)} dB_u | F_s]$$

$$= 0.04 + e^{-0.8t} (r_0 - 0.04) + E[\int_0^t 0.1e^{0.8(u-t)} dB_u | F_s] \quad [1]$$

$$= 0.04 + e^{-0.8t} (r_0 - 0.04) + E[\int_0^s 0.1e^{0.8(u-t)} dB_u | F_s] + E[\int_s^t 0.1e^{0.8(u-t)} dB_u | F_s] \quad [1]$$

We have split the integral into two parts. In the first part from 0 to s , all random increments are known as we are conditioning on F_s . we can treat this integral as a constant and take it outside the expectation.

In the second integral from s to t , all increments are independent of the past and hence the conditioning on F_s can be dropped. The expectation of this integral will be zero since dB_u in itself has zero mean.

$$E[r_t | F_s] = 0.04 + e^{-0.8t} (r_0 - 0.04) + \int_0^s 0.1e^{0.8(u-t)} dB_u \quad [1]$$

From part (i), we can relate this to r_s

$$E[r_t | F_s] = 0.04 + e^{-0.8t} (r_0 - 0.04) + e^{-0.8(t-s)} \int_0^s 0.1e^{0.8(u-s)} dB_u \quad [1]$$

Using result from part (i),

$$E[r_t | F_s] = 0.04 + e^{-0.8t} (r_0 - 0.04) + e^{0.8(s-t)} [r_s - 0.04 - (r_0 - 0.04) e^{-0.8s}]$$

$$= 0.04 (1 - e^{0.8(s-t)}) + e^{0.8(s-t)} r_s$$

[1]

[5]

[14 Marks]

Solution 8:

- i) The EMH (in its various forms) states that asset prices reflect information. However it does not explicitly tell us how new information affects prices. It is also empirically difficult to establish precisely when information arrives — for example, many events are widely rumoured prior to official announcements.

[1]

Many studies show that the market over-reacts to certain events and underreacts to other events. The over/under-reaction is corrected over a long time period. If this is true then traders could take advantage of the slow correction of the market, and efficiency would not hold.

[2]

[3]

ii)

- Past performance predictability: past winners subsequently underperform
- Accounting ratios have predictive power: e.g. P/E predicts low future returns
- Firms coming to market have poor subsequent performance.
- Stock prices take some time to react to earnings announcements
- Abnormally poor performance following mergers
- Abnormally good performance following demergers.

[3]

[6 Marks]

Solution 9:

- i) If the salesperson prefers more to less wealth, then $U'(w) > 0$. Since $U(w) = w - 0.3w^2$, we have $U'(w) = 1 - 0.6w > 0$ or $w < 1.67$.

He is also risk-averse for all $w > 0$ which follows from $U''(w) = -0.6 < 0$. So no extra constraints on the range of w follow, and the salary range for which the utility function is appropriate will be up to 1.67.

[2]

- ii) His expected remuneration will be $0.60(1000,000) + 0.40 \times 70,000 = \text{Rs.}88000$

$$\text{His expected utility} = 0.60(1 - 0.3 \times 1^2) + 0.40(0.7 - 0.3 \times 0.7^2)$$

$$= 0.6412$$

[2]

- iii) The minimum level of fixed salary will be x (in Rs 100,000s), the certainty equivalent, where

$$x - 0.3x^2 = 0.6412 \Rightarrow 0.3x^2 - x + 0.6412 = 0 \Rightarrow x = 2.46 \text{ or } 0.867$$

2.33 is outside the appropriate range of w , so we can ignore this value. Hence, the minimum will be 0.867 or Rs 86,700.

[3]

[7 Marks]

Solution 10:

i)

- $\psi(U)$ does not depend on λ . This parameter affects the speed with which the process runs, but does not affect the ultimate probability of ruin.

- $\psi(U)$ is higher for higher values of μ since the significance of the starting capital falls as μ rises, providing proportionately less of a buffer.
- $\psi(U)$ is lower for higher values of θ since the higher θ is the higher the premiums with no change to claim amounts, so that there is a larger buffer against ruin.
- $\psi(U)$ is lower for higher values of U since the higher U is the higher the larger the buffer against ruin given by the initial capital.

[4]

ii) The answer is b.

[2]

[6 Marks]

Solution 11:

Incremental claims in mid-2018 prices:

Accident year	Development year		
	0	1	2
2016	1762.88	677.33	230.00
2017	1941.33	942.00	
2018	2012.00		

[2]

Cumulative claims in mid-2018 prices:

Accident year	Development year		
	0	1	2
2016	1762.88	2440.21	2670.21
2017	1941.33	2883.33	
2018	2012.00		

[1]

DF 0,1	$=(2440.21+2883.33)/(1762.88+1941.33)=1.437$
DF 1,2	$=2670.21/2440.21$ $= 1.094$

[1]

Completed Cumulative claims:

Accident year	Development year		
	0	1	2
2016			
2017			3155.10
2018		2891.57	3164.11

[1]

Incremental claim (mid 2018 prices):

Accident year	Development year		
	0	1	2
2016			
2017			271.77
2018		879.57	272.54

[2]

Projecting for inflation

$$= (879.57+271.77)*115/112+272.54*118/112= 1469.31$$

[1]

[8 Marks]
